Class 9 Mathematics — Exercise 6.2

Basic Concept

Two formulae of this exercise:

$$\boxed{l = r\theta} \qquad \boxed{A = \frac{1}{2}r^2\theta}$$

where l = length of circular arc, r = radius of circle, θ = central angle **in radians**, A = area of sector.

(i) Arc: A part of the circumference of a circle is called an arc.

(ii) Segment of Circle: A part of the circle bounded by an arc and a chord is called segment of circle.

(iii) Sector of Circle: A part of the circle bounded by the two radii and an arc is called sector of circle.

Conversion of angles:

$$\text{Degree} \xrightarrow{\times \frac{\pi}{180}} \text{Radian} \qquad \text{Radian} \xrightarrow{\times \frac{180}{\pi}} \text{Degree}$$

In this exercise we take $\pi = 3.1416$.

Proof: Length of an arc, $l = r\theta$

Length of a circular arc and radian measure of its central angle are directly proportional. Let θ be the central angle of an arc of length l, and let l_1 be the length of an arc whose central angle is 1 radian, so $l_1 = r$.

$$l : \theta :: l_1 : 1$$

$$l : \theta :: r : 1$$

Product of extremes = Product of means:

$$l \times 1 = r \times \theta$$

$$\boxed{l = r\theta}$$

So, length of a circular arc is the product of radius and radian measure of central angle.

Proof: Area of sector, $A = \frac{1}{2}r^2\theta$

Area of sector is directly proportional to the radian measure of its central angle:

$$\text{area of sector} : \theta :: \text{area of circle} : \text{complete angle}$$

$$A : \theta :: \pi r^2 : 2\pi$$

Product of extremes = Product of means:

$$2\pi A = \pi r^2 \theta$$

$$A = \frac{\pi r^2 \theta}{2\pi}$$

$$\boxed{A = \frac{1}{2}r^2\theta}$$

where θ is in radians.

Question 1

Find length of arc and area of sector for given radius and central angle.

(i) $r = 5$ cm, $\theta = \frac{\pi}{3}$ radians

Length of arc $= l =$? Area of sector $= A =$?

Length of arc:

$$l = r\theta = (5)\left(\frac{\pi}{3}\right) = \frac{5(3.1416)}{3}$$

$$\boxed{l = 5.24 \text{ cm}}$$

Area of sector:

$$A = \frac{1}{2}r^2\theta = \frac{1}{2}(5)^2 \times \frac{\pi}{3} = \frac{1}{2} \times 25 \times \frac{3.1416}{3} = \frac{78.54}{6}$$

$$\boxed{A = 13.09 \text{ cm}^2}$$

(ii) $r = 12$ m, $\theta = 120^\circ$

Length of arc $= l =$? Area of sector $= A =$?

First, we convert angle from degree to radian:

$$\theta = 120^\circ = 120 \times \frac{\pi}{180} = \frac{2\pi}{3}$$

Length of arc:

$$l = r\theta = (12)\left(\frac{2\pi}{3}\right) = \frac{24(3.1416)}{3} = \frac{75.3984}{3}$$

$$\boxed{l = 25.13 \text{ m}}$$

Area of sector:

$$A = \frac{1}{2}r^2\theta = \frac{1}{2}(12)^2 \times \frac{2\pi}{3} = \frac{1}{2} \times 144 \times \frac{2(3.1416)}{3} = \frac{452.3904}{3}$$

$$\boxed{A = 150.80 \text{ m}^2}$$

(iii) $r = 6$ **dm,** $\theta = 60^\circ\, 45'\, 30''$

Length of arc $= l = ?$ Area of sector $= A = ?$

First, we convert angle from degree to radian:

$$\theta = 60^\circ\, 45'\, 30'' = 60^\circ + 45' + 30''$$

$$= 60^\circ + \left(\frac{45}{60}\right)^\circ + \left(\frac{30}{60 \times 60}\right)^\circ$$

$$= 60^\circ + 0.75^\circ + 0.0083^\circ = 60.7583^\circ$$

$$\theta = 60.7583 \times \frac{\pi}{180} = 60.7583 \times \frac{3.1416}{180} = 1.0604 \text{ rad}$$

Length of arc:

$$l = r\theta = (6)(1.0604)$$

$$\boxed{l = 6.36 \text{ dm}}$$

Area of sector:

$$A = \frac{1}{2}r^2\theta = \frac{1}{2}(6)^2(1.0604) = \frac{1}{2}(36)(1.0604)$$

$$\boxed{A = 19.09 \text{ dm}^2}$$

Question 2

A central angle in a circle of radius 4 cm is 75°. Find the length of the intercepted arc and area bounded by the sector.

$r = 4$ cm, $\theta = 75^\circ$

Length of arc $= l = ?$ Area of sector $= A = ?$

First, we convert angle from degree to radian:

$$\theta = 75^\circ = 75 \times \frac{\pi}{180} = 75 \times \frac{3.1416}{180} = 1.309 \text{ rad}$$

Length of arc:

$$l = r\theta = (4)(1.309)$$

$$\boxed{l = 5.24 \text{ cm}}$$

Area of sector:

$$A = \frac{1}{2}r^2\theta = \frac{1}{2}(4)^2(1.309) = \frac{1}{2}(16)(1.309)$$

$$\boxed{A = 10.47 \text{ cm}^2}$$

Question 3

Find radian measure of the central angle of a circular sector for the following data.

(i) $l = 8$ **cm**, $r = 4$ **cm**

$\theta = ?$

From $l = r\theta$: $\quad \frac{l}{r} = \theta \Rightarrow \boxed{\theta = \frac{l}{r}}$

$$\theta = \frac{l}{r} = \frac{8}{4}$$

$$\boxed{\theta = 2 \text{ rad}}$$

(ii) $l = 8.5$ **m**, $r = 2.25$ **m**

$\theta = ?$

$$\theta = \frac{l}{r} = \frac{8.5}{2.25}$$

$$\boxed{\theta = 3.78 \text{ rad}}$$

(iii) $A = 45$ **cm**^{2}, $r = 10.70$ **cm**

$\theta = ?$

From $A = \frac{1}{2}r^2\theta$: $\quad \frac{2A}{r^2} = \theta \Rightarrow \boxed{\theta = \frac{2A}{r^2}}$

$$\theta = \frac{2A}{r^2} = \frac{2(45)}{(10.70)^2} = \frac{90}{114.49}$$

$$\boxed{\theta = 0.79 \text{ rad}}$$

(iv) $A = 100$ **cm**^{2}, $l = 10$ **cm**

$\theta = ?$

From $l = r\theta$: $\quad \boxed{r = \frac{l}{\theta}}$

Put the value of r in the formula of area:

$$A = \frac{1}{2}r^2\theta = \frac{1}{2} \times \frac{l^2}{\theta^2} \times \theta$$

$$A = \frac{l^2}{2\theta} \;\Rightarrow\; \theta = \frac{l^2}{2A}$$

$$\theta = \frac{(10)^2}{2(100)} = \frac{100}{2(100)} = \frac{1}{2}$$

$$\boxed{\theta = 0.5 \text{ rad}}$$

Question 4

Find radius of circle for the following information.

(i) $l = 4$ cm, $\theta = \pi$ radians

$r = ?$

From $l = r\theta$: $\boxed{r = \frac{l}{\theta}}$

$$r = \frac{l}{\theta} = \frac{4}{\pi} = \frac{4}{3.1416}$$

$$\boxed{r = 1.27 \text{ cm}}$$

(ii) $l = 6$ m, $\theta = 15^\circ$

$r = ?$

First, we convert angle from degree to radian:

$$\theta = 15^\circ = 15 \times \frac{\pi}{180} = 15 \times \frac{3.1416}{180} = 0.2618 \text{ rad}$$

$$r = \frac{l}{\theta} = \frac{6}{0.2618}$$

$$\boxed{r = 22.92 \text{ m}}$$

(iii) $A = 200$ cm^2, $\theta = \frac{\pi}{4}$ radians

$r = ?$

$$\theta = \frac{\pi}{4} = \frac{3.1416}{4} = 0.7854 \text{ rad}$$

From $A = \frac{1}{2}r^2\theta$: $\frac{2A}{\theta} = r^2 \;\Rightarrow\; \sqrt{\frac{2A}{\theta}} = \sqrt{r^2} \;\Rightarrow\; \boxed{r = \sqrt{\frac{2A}{\theta}}}$

$$r = \sqrt{\frac{2A}{\theta}} = \sqrt{\frac{2(200)}{0.7854}} = \sqrt{509.294}$$

$$\boxed{r = 22.57 \text{ cm}}$$

(iv) $A = 100\ \mathbf{dm}^2,\ l = 10\ \mathbf{dm}$

$r = ?$

From $l = r\theta$: $\quad \dfrac{l}{r} = \theta$

Put the value of θ in the formula of area:

$$A = \frac{1}{2}r^2\theta = \frac{1}{2}r^2 \times \frac{l}{r}$$

$$A = \frac{lr}{2} \ \Rightarrow\ \frac{2A}{l} = r \ \Rightarrow\ \boxed{r = \frac{2A}{l}}$$

$$r = \frac{2(100)}{10} = \frac{200}{10}$$

$$\boxed{r = 20 \text{ dm}}$$

Question 5

A 30 inch pendulum swings through an angle of 30°. Find the length of the arc in inches through which the tip of the pendulum swings.

$r = 30$ inch, $\quad \theta = 30°, \quad l = ?$

First, we convert angle from degree to radian:

$$\theta = 30° = 30 \times \frac{\pi}{180} = 30 \times \frac{3.1416}{180} = 0.5236 \text{ rad}$$

$$l = r\theta = (30)(0.5236)$$

$$\boxed{l = 15.71 \text{ inch}}$$

Question 6

A motorcycle is traveling on a curve along a highway. The curve is an arc of a circle with radius of 10 km. If the motorcycle's speed is 42 km/h, what is the angle in degrees through which the motorcycle will turn in 21 minutes?

$r = 10$ km

$v = 42$ km/h $= \dfrac{42}{60}$ km/min $= 0.7$ km/min

$t = 21$ min, $\quad \theta° = ?$

Distance covered (length of arc), using $S = vt$:

$$l = (0.7)(21)$$

$$l = 14.7 \text{ km}$$

From $l = r\theta$: $\quad \dfrac{l}{r} = \theta \ \Rightarrow\ \theta = \dfrac{l}{r}$

$$\theta = \frac{14.7}{10} = 1.47 \text{ rad}$$

Now convert radian into degree:

$$\theta = 1.47 \times \frac{180}{\pi} = 1.47 \times \frac{180}{3.1416} = 84.22^\circ$$

$$\boxed{\theta \approx 84^\circ}$$

Question 7

Find perimeter and area of a half circle in the following figure (diameter AB = 12 cm) by using the formulae $l = r\theta,\ A = \frac{1}{2}r^2\theta$.

Perimeter of half circle $= P = ?$

Area of half circle $= A = ?$

$$r = \frac{d}{2} = \frac{12}{2} = 6 \text{ cm}$$

For a half circle the central angle is $\theta = 180^\circ = \pi$ rad.

Perimeter: (curved arc + diameter)

$$P = l + d = r\theta + d$$

$$= (6)\pi + 12 = 6(3.1416) + 12$$

$$= 18.8496 + 12$$

$$\boxed{P = 30.85 \text{ cm}}$$

Area:

$$A = \frac{1}{2}r^2\theta = \frac{1}{2}(6)^2(\pi) = \frac{1}{2}(36)(3.1416) = 18(3.1416)$$

$$\boxed{A = 56.55 \text{ cm}^2}$$

Question 8

Find the circular measure of the angle between the hour hand and minute hand of a clock at:

(i) 9 o'clock **(ii)** 02:30 **(iii)** 06:45

1st Method

(i) 9 o'clock

A clock is like a circle.

Total angle $= 360^\circ$

Total parts (hours) $= 12$

$$\text{Angle for one part} = \frac{360^\circ}{12} = 30^\circ$$

At 9 o'clock there are 3 parts between the hour hand (at 9) and the minute hand (at 12):

$$\text{Angle b/w hour hand and minute hand} = 3 \times 30^\circ = 90^\circ$$

Now, we convert angle from degree to radian:

$$= 90 \times \frac{\pi}{180}$$

$$\boxed{\text{Angle} = \frac{\pi}{2} \text{ rad}}$$

2nd Method

Hour hand:

$$1 \text{ rotation} = 360^\circ, \quad 12 \text{ h} = 360^\circ, \quad 1 \text{ h} = \frac{360^\circ}{12} \Rightarrow \boxed{1 \text{ h} = 30^\circ}$$

Minute hand:

$$1 \text{ rotation} = 360^\circ, \quad 60 \text{ min} = 360^\circ, \quad 1 \text{ min} = \frac{360^\circ}{60} \Rightarrow \boxed{1 \text{ min} = 6^\circ}$$

Formulae (angles measured from 12):

$$\text{Hour hand} = \left(\text{Hours} + \frac{\text{Min}}{60}\right) 30^\circ \qquad \text{Minute hand} = \text{Min} \times 6^\circ$$

$$\text{Angle between them} = |\,\text{Minute hand} - \text{Hour hand}\,|$$

(i) 9 o'clock $\rightarrow$ 09:00 $\Rightarrow$ Hours = 9, Minutes = 0

Hour hand:

$$\left(\text{Hours} + \frac{\text{Min}}{60}\right) 30^\circ = \left(9 + \frac{0}{60}\right) 30^\circ = (9)30^\circ = 270^\circ$$

Minute hand: at 12, i.e. 0° or 360°.

Angle b/w hour hand and minute hand (at 9 o'clock):

$$\text{Angle} = |\,360^\circ - 270^\circ\,| = 90^\circ = 90 \times \frac{\pi}{180}$$

$$\boxed{\text{Angle} = \frac{\pi}{2} \text{ rad}}$$

(ii) 02:30 $\Rightarrow$ Hours = 2, Minutes = 30

Hour hand:

$$\left(\text{Hours} + \frac{\text{Min}}{60}\right) 30^\circ = \left(2 + \frac{30}{60}\right) 30^\circ = (2 + 0.5)30^\circ = (2.5)30^\circ = 75^\circ$$

Minute hand:

$$\text{Min} \times 6^\circ = 30 \times 6^\circ = 180^\circ$$

Angle b/w hour hand and minute hand (at 2:30):

$$\text{Angle} = |\, 180^\circ - 75^\circ \,| = 105^\circ = 105 \times \frac{\pi}{180}$$

$$\boxed{\text{Angle} = \frac{7\pi}{12} \text{ rad}}$$

(iii) 06:45 $\Rightarrow$ Hours = 6, Minutes = 45

Hour hand:

$$\left(\text{Hours} + \frac{\text{Min}}{60}\right) 30^\circ = \left(6 + \frac{45}{60}\right) 30^\circ = (6 + 0.75)30^\circ = (6.75)30^\circ = 202.5^\circ$$

Minute hand:

$$\text{Min} \times 6^\circ = 45 \times 6^\circ = 270^\circ$$

Angle b/w hour hand and minute hand (at 6:45):

$$\text{Angle} = |\, 270^\circ - 202.5^\circ \,| = 67.5^\circ = 67.5 \times \frac{\pi}{180}$$

$$\boxed{\text{Angle} = \frac{3\pi}{8} \text{ rad}}$$